

Ravinder Jilkapally

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Professional Summary

1. Engineering Leader with 18+ years of experience across Web, Data, AI/ML platforms, agentic systems, large-scale backend architectures, and multi-cloud MLOps.
2. Expert in taking products from 0→1, 1→N scaling, and building world-class high-performing engineering organizations.
3. Blend of hands-on technical leadership (distributed systems, multimodal inference, PyTorch, vector DBs, RAG, LangGraph, Bedrock/SageMaker, GPU optimization) and executive execution discipline (SLOs, OKRs, platform strategy, security, compliance, cost governance, ROI-based prioritization).
4. Experience spanning Generative AI, Data, Analytics, MLOps, and cloud-scale architecture across regulated industries, including mortgage and financial services.
5. Successfully launched AI products at Freedom Mortgage, CHL, Village Capital, and Mortgage Connect, supporting Sales, Servicing, and HMDA compliance.
6. Building document intelligence systems at Xpanse AI using multimodal LLMs (Claude, Gemini, Gemma 4, DeepSeek), agentic pipelines, and research tooling for mortgage document classification. Increased model accuracy through fine-tuning open models (Gemma 4, DeepSeek), moving away from legacy InternVL models. Two-time NVIDIA hackathon finalist (1st place DGX AITX 2025, runner-up GTC 2026 showcased on NVIDIA Developer Live).
7. Deep experience integrating Snowflake Cortex, Vector Search, Document AI, and LLM Functions into operational pipelines for real-time analytics, reasoning automation, and cross-system decisioning.
8. Led architecture, productization, and deployment of multi-tenant AI platforms, across call summarization, document intelligence (800+ document types), and compliance assistants.
9. Partnered with Cenlar to deliver an end-to-end GenAI Platform, integrating call and document workflows with scalable cloud-native AI/ML infrastructure.
10. Previously led the data platform modernization at Rocket Mortgage, building enterprise analytics and ML pipelines using AWS and Cloudera, enabling scalable data science workflows.
11. Established MLOps, DevOps, and SDLC best practices, ensuring seamless transition of data science models from notebooks to production environments with auditability and governance.
12. Led cross-functional teams across engineering, ML, and product; enabled rapid iteration and experimentation while maintaining production-grade delivery and stakeholder trust.
13. Worked closely with executive leadership, legal, security, and business functions to deliver GenAI solutions within strict compliance boundaries (e.g., SOC2, NPPI, HMDA).
14. Act as an advisor and architect for early-stage GenAI startups, helping shape product vision, build v1 teams, and deliver POCs across industries.
15. Passionate about turning AI into operational leverage, building scalable platforms, and enabling organizations to harness the power of data and intelligent automation.

Education

Master of Science in Computer Science

2008

Wayne State University, Detroit, MI

Bachelor of Technology in Information Technology

2006

Jawaharlal Nehru Technological University, Hyderabad, India

Professional Experience

Director, GenAI Product & Engineering – Xpanse, Remote

May 2024 – Present

Lead end-to-end Generative AI engineering, delivering multi-product GenAI solutions at enterprise scale and architecting the company's secure multi-tenant AI platform.

Platform Architecture & 0→1 AI Systems

- Designed and built a **multi-tenant GenAI platform** supporting text, audio, document, and metadata pipelines with **RAG, vector search, semantic indexing, model routing, and multi-agent orchestration**.
- Built **Agentic IDP Experiments Platform** (Next.js 14 + Prisma + PostgreSQL) used daily by the IDP team: AI Studio with multi-model comparison (Claude Sonnet/Haiku, Gemini Pro/Flash), Triage Workbench, Agentic Review, Prompt Browser, and multi-model experiment tracking — accelerating document extraction accuracy across model families.
- Architected and shipped **VCAP Intelligence Platform**: production-grade dashboard for multimodal document classification pipeline with model leaderboard, real-time run polling, side-by-side run comparison with per-document accuracy deltas, and 5-stage pipeline visualization (PDF render → VLM classification → signal computation → boundary detection → Bedrock eval).
- Architected hybrid cloud + on-prem patterns enabling **air-gapped, SOC2-aligned, NPPI-compliant deployments**.
- Led development of **VLM-based document classification research pipeline**: Migrated from legacy InternVL3-2B to fine-tuned open models (Gemma 4, DeepSeek) for per-page classification, improving accuracy for Village Capital document workflows. Built 6-signal weighted boundary detection and an autoresearch loop that automates prompt optimization against ground truth — closing the feedback loop from classification failures to promoted prompt versions without manual intervention. Adapted pipeline for agentic workflows with tool-use orchestration and autonomous document routing.

AI Product Delivery & Scale

- **Call Intelligence Platform: Reduced review time from 90 min to <5 min, providing structured insights (intent, adherence, actions)**. Scaled to **2M+ minutes per month**.
- **Document Intelligence**: Delivered **multimodal LLM-based document workflows** (Claude, Gemini, Gemma 4, DeepSeek) across 800+ document types with **90%+ accuracy**, processing **1M+ pages/month**. Integrated schema-anchored extraction with 419+ business rule validations, party name matching (fuzzy + NLP), and per-field coverage diagnostics.
- **AI Workspace Studio**: Built secure enterprise chat + knowledge-grounding, model switching, vector search, and multi-file reasoning environment.
- Implemented **Snowflake Cortex integration**, enabling real-time insights, vector search, and LLM-driven analytics directly within Snowflake for Ops, Compliance, and Sales teams.

Team Building & Leadership

- Scaled and managed a 45+ person engineering organization across SWE, ML, DS, QA, DevOps, and PM (US + offshore).
- Created career ladders, engineering standards, hiring pipelines, and technical leadership structures.
- Drove weekly engineering reviews, architectural governance, roadmap planning, and executive alignment.

Operational Excellence & Governance

- Defined CI/CD, MLOps, LLMOps, evals, guardrails, prompt governance, traceability, and multi-tenant release processes.
- Delivered AI within strict regulatory, audit, NPPI, and SOC2 requirements.
- Standardized deployment automation reducing client onboarding cycle from weeks to days.

Principal Consultant – AISOFT LLC, Austin, TX | aisoft.us**Jan 2022 – Present**

- Founded AISOFT LLC to provide Web, GenAI, data, and AI/ML strategy consulting for early-stage startups and enterprises navigating digital transformation and AI adoption.
- Advised multiple startups in architecture planning, team formation, and technology roadmap definition — enabling smooth transitions from concept to MVP.
- Led development and rollout of custom AI and analytics solutions, including early POCs for call summarization, document intelligence, and data compliance.
- Served as a technical partner to founding teams, helping evaluate feasibility, set up infrastructure, and align product development with go-to-market strategies.
- Worked with GenAI accelerators and startup ecosystems to explore investment and incubation opportunities.
- Actively exploring partnerships to scale AISOFT's capabilities in the US and India, focusing on high-impact, niche GenAI use cases in regulated domains.

Products & Applied AI (2025–2026)

- **[RefereAI](#)** – AI sports referee using NVIDIA Cosmos Reason 2 (8B) for physical chain-of-thought reasoning over sports video. Five sports, 3 analysis modes (scene, physics, commentary), broadcast-style 3-panel UI. Runs on Jetson AGX Orin. NVIDIA Cosmos Cookoff.
- **[Sideline](#)** – Modular AI sports robot combining Qwen vision-language model with confidence-threshold referee fallback ($\geq 0.75 \rightarrow$ autonomous call; $< 0.75 \rightarrow$ VAR rectangle for human review). Three robot form factors: SO-101 arm (LeRobot), MentorPi (ROS2), Unitree G1 humanoid. 4-person team, 6-hour build. Nebius.Build SF, March 2026.
- **[CoachClaw](#)** – On-device AI sideline assistant for volunteer coaches running on NVIDIA GB10 with zero cloud dependency. Integrated 259K crime incidents + 21K food inspection records for local safety context. Delivered via Telegram bot — no app install required. GTC 2026 Hack for Impact.
- **[Studio Copilot](#)** – Local-inference AI workspace for photographers replacing 5 tools (curation, client review, contracts, invoices, follow-ups). OpenCLIP ViT-B/32 pipeline on NVIDIA GB10: 182 photos $\rightarrow$ 26 clusters $\rightarrow$ 43 highlights in ~ 2 s. Runner-up, GTC 2026 Hack to Create; showcased on NVIDIA Developer Live. 90 GitHub cloners within 24 hours.
- **[Undervolt](#)** – GPU-accelerated urban intelligence platform analyzing 2.3M+ Austin construction permits. NLP pipeline surfaced a 22:1 solar-to-battery ratio (unknown to city planners) and a $12\times$ equity gap in backup power across districts. Stack: NVIDIA Nemotron via NIM, RAPIDS, Supabase, React. 1st place, Urban Growth – NVIDIA DGX Spark Frontier Hackathon.
- **[HD Research Hub](#)** – Autonomous AI agent for Huntington's Disease research: literature analysis, drug repurposing exploration, and clinical trial tracking via PubMed, ClinicalTrials.gov, and Open Targets APIs.
- **[StudyPal](#)** – Multimodal AI homework assistant with real-time voice tutoring via Gemini Live API, PDF analysis, and Schoology auto-sync for K–8 students. Built for the Gemini Live Agent Challenge.
- **[Applly](#)** – AI-powered job search assistant that analyzes resume-job fit, generates tailored cover letters, cold emails, and interview prep using Gemini 1.5 Flash. Austin AI Hackathon, March 2025.

Engineering and Architecture Lead – Rocket Mortgage, Detroit, MI**Jul 2020 – Nov 2023**

- Successfully led and evangelized pragmatic Data Mesh initiatives at Rocket Central, resulting in an increase in data product adoption across the organization and led to 50% increase in our Data Product adoption.
- Lead, design, build and create a trusted ecosystem of Data Products that span ML, Enrichment, and Analytical areas for Rocket Central and the family of companies within Rocket Companies.
- Delivered engineering and architecture foundational artifacts for over 20+ Data Product development projects, resulting in a 75% reduction in development time.
- Delivered successful Data and Platform strategies driven by Company OKRs, achieving a 100% success rate.

- Built highly effective teams across Data Engineering, Platform, and Infrastructure, resulting in a 50% increase in team collaboration and coordination.
- Partnered with cross-functional teams to deliver well-rounded solutions, resulting in a 20% increase in project efficiency and productivity.
- Managed, mentored, and coached a team of sizes 5 – 15 that included Software and Machine Learning Engineers, Data Scientists, Solution and Data Architects which resulted in a 50% increase in team performance and 90% favorability in Team surveys.
- Provided technical direction, strategy, and roadmaps for initiatives across Data Engineering, Platform, and Infrastructure, resulting in a 15% increase in project success rate.
- Established and led practices for Data as First Class Citizen, Data Governance (Metadata, Data Quality, Data Classification, Data Privacy, MDM), and Data Observability leveraging DevSecOps, resulting in a 75% reduction in data-related risks and incidents.
- Provide technical consulting for companies under Rocket Companies (Rocket Mortgage, Rocket Homes, Rocket Auto, Amrock), including knowledge sharing and expertise in Data, Platform, and Infrastructure areas, which accelerated their Cloud adoption by 80%.
- Ensured 100% Data Platform compliance for data security and privacy and develop risk mitigation strategies.
- Created new standards and documented decisions for the organization's Data Platform, resulting in a 25% increase in data quality, consistency, and enterprise adoption.
- Evaluated and introduced new technologies, frameworks, and products in the Data and Platform space, resulting in a 15% increase in project efficiency and innovation.
- Delivered performance reviews, reviewed individual OKRs, and provided compensation adjustments to support and recognize team members, resulting in a 75% increase in employee retention and satisfaction.

Big Data Architect – Rocket Mortgage(Quicken Loans), Detroit, MI Jan. 2019 – Jul 2020

- Building software integrations for Data Governance, Data Lake, Data Science, and Data Product applications.
- Design and build a Software Framework for helping build, design, analyze and visualize Data Science and Data Product Applications with a focus on containers.
- Design and build software libraries to help with seamless integrations with Big Data Platform Services – Kafka, HBase, Hive, Oozie, and AWS Services like DynamoDB, Elastic Search, RedShift, etc
- Provide technical direction, planning, and roadmaps for initiatives around Data Science, Data Products, and Data Governance applications.
- Led several Data Science and Data Product applications from Design to launch.
- Building rapid PoCs and evaluating new technologies such as StreamAnalytix, Kinetica, and Kafka Enterprise Bus.
- Mentor teams for building applications with best software engineering practices, including developing expertise around Logging, Monitoring, and Alerting.
- Build and run workshops to train Team Members on using software frameworks and building reusable and configuration-driven applications.
- Tools and Technologies used – AWS, Cloudera, Collibra, Apache Kafka, Apache Spark, Python, Java, Apache Hive, Apache Atlas, CircleCI.

Senior Software Engineer – Rocket Mortgage, Detroit, MI Oct. 2017 – Jan. 2019

- Building integration software between Data Governance, Data Lake, and Data Science applications.
- Build, Design, and Develop real-time streaming applications, Machine Learning Model Execution framework.
- Tools and Technologies used – AWS, Collibra, Mule, Hortonworks HDP/HDF, Apache Spark, Python, Java, Apache Hive, Apache Atlas, CircleCI.
- Implement CI/CD pipelines for Machine Learning applications.

Senior Software Engineer – Dynatrace LLC, (Compuware), Detroit, MI Aug. 2014 – Oct. 2017

- Perform new product R&D using Data Analytics, Collective Intelligence, and Machine Learning techniques
- Building large-scale real-time analytics platform to extract data detect anomalies, classification, pattern recognition, generate real-time alerts, create visualizations, and provide recommendations using Apache Kafka, Cassandra, MongoDB, Apache Hadoop, Java, Node.js, D3, C3, Apache Spark, and Apache Mahout.
- Generate data-driven infographics and visualizations for showcasing product features and capabilities.
- Won ‘Employee of the Quarter’ for integrating Big Data Analytics results into Dynatrace products.
- Conceptualized, designed, and developed Outage Analyzer, a tool for tracking internet service outages.
- Conceptualized and designed a ‘Year in Outages’ infographic featured on the Dynatrace homepage.
- Introduced Continuous Integration and automated testing through Jenkins, Cucumber, Spock, JUnit, and Gradle.
- Implemented centralized logging through Elasticsearch, Logstash, and Kibana stack (ELK) for servers and applications deployed in Amazon EC2, which decreased the time to find issues in logs and thereby improved developer productivity.
- Improved productivity of internal benchmark publishing tools, and increased developer productivity by 70%.

Software Engineer – Compuware Corporation, Detroit, MI Jul. 2011 – Aug. 2014

- Part of an Intrapreneurial group focused on Big Data Analytics for Dynatrace’s Application Performance Management products.
- Design, prototype, and refine software application solutions to meet customer requirements.
- Work closely with designers, planners, testers, and marketing to envision, Design, and prototype web applications.
- Perform data analysis and mining on large-scale data sets stored in Apache Hadoop, Cassandra, MongoDB, MySQL, Solr, and Memcached using technologies such as Java, Groovy, RDF, Sesame, RabbitMQ, Spring, Hibernate, and Grails.
- Build REST APIs using tools and frameworks such as Java, Groovy on Grails, Ruby on Rails, and Git on Amazon EC2.
- Provide technical documentation and interact with Quality Assurance teams to provide results and reports.

Programmer Analyst – IDEACore LLC, Birmingham, MI Oct. 2008 – Jul. 2011

- Projects included web/desktop application development, data analysis, reporting, document, and content management.
- Set up development and production web servers and databases for J2EE, .NET, and Ruby on Rails applications using Apache HTTP/Tomcat web servers and MySQL/Oracle 10g database.
- Improved scalability and performance and development productivity of Java based applications using the latest tools and frameworks such as Groovy, Grails and, Apache Lucene, SOLR.
- Introduced Semantic web technologies such as ontologies, RDF, and OWL for building an e-commerce product recommendation system patented to IDEACore.

Skills

Languages & Scripting

Python, Java, SQL, C++, R, JavaScript, Bash, Groovy, C#, Ruby

Cloud & Infrastructure

AWS (S3, Lambda, Bedrock, ECS Fargate, SageMaker, DynamoDB), Azure (OpenAI, Entra ID), GCP (Vertex AI), Terraform, Kubernetes, Docker, CloudFormation, Snowflake, Cortex LLM Functions, Snowflake Vector Search, Snowpark, external functions for LLM orchestration

Integrations

API platform design, REST/gRPC, SDK development, event-driven services, Kafka, RabbitMQ, API orchestration, authentication/authorization workflows.

Data Platforms & Storage

Snowflake, Redshift, Postgres, MySQL, MongoDB, DynamoDB, HBase, Cassandra, Elasticsearch, Neo4j, Redis, HDFS

Big Data & Streaming

Kafka, Spark, Hadoop, Hive, Presto, Athena, Cloudera, Zookeeper, RabbitMQ

ML & GenAI

PyTorch, Hugging Face, MLflow, LangChain, LangGraph, LlamaIndex, Ollama, LMStudio, Jan, ChromaDB, Pinecone, FAISS, RAG architectures, embeddings, multi-agent orchestration, LLM fine-tuning (LoRA, QLoRA), SFT, prompt engineering, Hallucination reduction, SageMaker, Bedrock Claude, OpenAI, Gemini, GPT-4o, Agentic AI, LangGraph, tool-use orchestration, planning agents, retrieval agents, workflow agents, autonomous document routing agents, NVIDIA Cosmos Reason 2, Gemma 4, DeepSeek, Qwen, Nemotron, Jetson AGX Orin, DGX Spark/GB10 (Blackwell), MuJoCo, OpenCLIP ViT-B/32, Edge AI inference, llama.cpp GGUF, OpenClaw, MCP (Model Context Protocol), A2A patterns, open model fine-tuning (LoRA/QLoRA)

MLOps & DevOps

MLflow, GitHub Actions, Jenkins, CircleCI, Containerized model deployment, CI/CD, feature stores, model registries, CloudWatch, ELK Stack, monitoring, tracing, evaluations (Evals), Palantir AIP, agent monitoring, governance frameworks

Frameworks & APIs

FastAPI, Flask, Spring Boot, Node.js, .NET, React, Next.js 14 App Router, TypeScript, Tailwind CSS, Prisma, Recharts, REST, GraphQL, WebSockets, Event-driven architectures

Data Governance & Compliance

Collibra, Apache Atlas, Data lineage, metadata management, data quality, MDM, NPPI/PII controls, SOC2, audit frameworks, secure LLM patterns

Visualization & Collaboration

Sigma, PowerBI, Tableau, Figma, Lucidchart, Jira, Confluence, Slack

Awards and Nominations

- Xpanse Employee Spotlight – [LinkedIn Post](#) - Xpanse, Remote
- Xpanse Way Award - Xpanse, Remote
- ISMs in Action Award – [We'll Figure It Out](#) - Rocket Companies, Detroit, MI
- Rock Honors Nomination – Rocket Mortgage, Detroit, MI
- Employee of the month – Rocket Mortgage, Detroit, MI
- Employee of the quarter – Compuware Corporation, Detroit, MI

Volunteering

- Code Michigan – www.codemichigan.org - Detroit, MI
- Detroit Java User Group – Co-Organizer – Detroit, MI
- GiveCamp – Coding for Charity – givecamp.org – Ann Arbor, MI
- International Open Data Hackathon – opendataday.org – Ann Arbor, MI
- Startup Weekend – startupweekend.org – Baltimore, MD